



BLUESTOCK MUTUAL FUND ANALYTICS

Capstone Project Report

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Technology Stack:

[Python, Pandas, NumPy, SQLite, Power BI, Plotly]

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Executive Summary

The Indian Mutual Fund industry has experienced significant growth over the last decade, driven by increasing investor awareness, digital investment platforms, and the rising popularity of Systematic Investment Plans (SIPs). As the number of schemes and investors continues to grow, it becomes increasingly important to evaluate fund performance, risk characteristics, investor behavior, and industry trends using data-driven techniques.

The Bluestock Mutual Fund Analytics project was developed to provide a comprehensive analytical framework for understanding the mutual fund ecosystem. The project integrates multiple datasets including fund master information, NAV history, investor transactions, portfolio holdings, benchmark indices, industry AUM data, and SIP inflow statistics. These datasets were processed using Python, Pandas, NumPy, and SQLite to build a robust analytics pipeline.

The project was divided into multiple phases including data ingestion, cleaning, exploratory data analysis, performance analytics, advanced analytics, and dashboard development. Performance evaluation metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were calculated to assess fund quality. Advanced risk metrics including Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, and Herfindahl-Hirschman Index (HHI) were also implemented.

A Power BI dashboard was developed to provide interactive visualizations covering industry overview, fund performance, investor analytics, and SIP market trends. In addition, a simple recommendation engine was built to suggest mutual funds based on investor risk appetite.

The results indicate that risk-adjusted performance metrics provide better fund evaluation than absolute returns alone. The project demonstrates how analytics can help investors make informed investment decisions and support data-driven fund management.

Problem Statement

The mutual fund industry offers investors access to a large number of investment schemes across equity, debt, hybrid, and sector-specific categories. While this variety provides flexibility, it also creates challenges in selecting suitable investment options. Most investors focus primarily on historical returns without considering important factors such as risk, volatility, portfolio concentration, and consistency of performance.

Traditional fund comparison methods often fail to provide a complete picture of a fund's risk-adjusted performance. Furthermore, investor behavior patterns such as SIP continuity, transaction frequency, and investment preferences are rarely analyzed in detail. Fund houses and investors therefore require a centralized analytics platform capable of combining multiple datasets and generating actionable insights.

The primary challenge addressed by this project is the development of a comprehensive mutual fund analytics framework that can evaluate performance, measure risk, analyze investor behavior, visualize industry trends, and provide fund recommendations based on investor risk profiles. The solution should support both descriptive and analytical insights while remaining scalable for future enhancements such as predictive analytics and live market integration.

Project Objectives

The primary objective of this project is to develop an end-to-end mutual fund analytics platform capable of processing large financial datasets and generating meaningful insights for investors and analysts.

The specific objectives of the project are:

1. To collect and integrate multiple mutual fund datasets including fund information, NAV history, investor transactions, portfolio holdings, benchmark indices, SIP inflows, and industry AUM data.
2. To design and implement an ETL (Extract, Transform, Load) pipeline for data cleaning, preprocessing, and storage in a structured SQLite database.
3. To perform Exploratory Data Analysis (EDA) and identify trends in fund performance, industry growth, investor participation, and asset allocation.
4. To calculate key performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown for evaluating mutual fund performance.
5. To perform advanced risk analytics including VaR, CVaR, Rolling Sharpe Ratio, SIP continuity analysis, investor cohort analysis, and sector concentration analysis.
6. To develop a Power BI dashboard providing interactive visualizations and decision-support tools.
7. To build a risk-based mutual fund recommendation engine capable of suggesting suitable funds according to investor risk appetite.
8. To generate actionable insights and recommendations that can assist investors in making informed financial decisions.

Data Sources

The project utilizes multiple datasets representing different aspects of the mutual fund ecosystem. Combining these datasets enables comprehensive analysis of fund performance, investor behavior, market trends, and portfolio characteristics.

The Fund Master dataset contains scheme-level information such as AMFI code, fund house, category, plan type, expense ratio, and risk grade. This dataset acts as the primary reference table used for integrating other datasets.

The NAV History dataset contains historical Net Asset Value (NAV) records for multiple mutual fund schemes. These records were used to calculate daily returns, CAGR, volatility, Sharpe Ratio, and other performance metrics.

The Investor Transactions dataset includes SIP, lumpsum, and redemption transactions performed by investors. This dataset was used for investor cohort analysis, SIP continuity analysis, transaction trend analysis, and demographic segmentation.

The Portfolio Holdings dataset contains stock-level holdings and sector allocations for mutual funds. It was used to evaluate diversification and calculate the Herfindahl-Hirschman Index (HHI).

The Industry AUM and SIP Inflow datasets provide insights into overall industry growth and retail participation trends. Benchmark indices such as Nifty 50 and Nifty 100 were used to compare mutual fund performance against broader market performance.

ETL Design and Data Architecture

Overview

The foundation of any analytics project is a reliable data pipeline. The Bluestock Mutual Fund Analytics project follows a structured ETL (Extract, Transform, Load) process to ensure that data collected from multiple sources is accurate, consistent, and ready for analysis.

The ETL pipeline was designed to handle mutual fund datasets containing scheme information, historical NAV records, investor transactions, portfolio holdings, benchmark data, industry AUM statistics, and SIP inflow records. These datasets were available in CSV format and required cleaning and standardization before analytical processing.

The ETL process ensures data quality, improves consistency across datasets, and enables efficient storage and retrieval using a centralized SQLite database.

Extract Phase

The first phase involved collecting raw datasets from various sources and storing them within the project structure. The datasets included:

- Fund Master dataset
- NAV History dataset
- Investor Transactions dataset
- Portfolio Holdings dataset
- Industry AUM dataset
- SIP Inflow dataset
- Benchmark Index dataset

These files were stored in the data/raw directory and served as the source data for the project.

The extraction process focused on validating file integrity, checking data availability, and understanding the structure of each dataset before transformation.

Transform Phase

The transformation phase involved cleaning and preparing the datasets for analysis.

The following operations were performed:

Data Cleaning

- Removal of duplicate records.
- Standardization of column names.
- Correction of inconsistent date formats.

- Handling of missing values.
- Verification of AMFI codes and fund identifiers.

Data Type Conversion

Several fields required conversion into appropriate formats:

- Date columns converted into datetime format.
- Numeric columns converted into float or integer types.
- Percentage fields standardized for analytical calculations.

Feature Engineering

Additional analytical fields were created:

- Daily Returns
- CAGR Metrics
- Sharpe Ratio
- Alpha and Beta
- Maximum Drawdown
- Fund Score
- Risk Categories

These derived features were later used for performance and risk analysis.

Data Integration

Multiple datasets were connected using common identifiers such as:

- AMFI Code
- Date
- Scheme Name

This integration enabled cross-dataset analysis and dashboard development.

Load Phase

After transformation, the cleaned datasets were stored in two formats:

Processed CSV Files

Cleaned datasets were saved inside the data/processed directory for future use and reproducibility.

SQLite Database

A centralized SQLite database named `bluestock_mf.db` was created to store the processed data.

Benefits of using SQLite:

- Lightweight database engine
- Easy integration with Python
- Fast querying for analytics
- Structured data storage

The database acted as a single source of truth for all project datasets.

Data Architecture

The project follows the architecture shown below:

Raw Datasets



Data Cleaning & Validation



Feature Engineering



Processed CSV Files



SQLite Database



Analytics & Metrics Calculation



Power BI Dashboard



Business Insights & Recommendations

This architecture ensured that all analyses were based on consistent, validated, and structured data.

Exploratory Data Analysis (EDA)

Introduction

Exploratory Data Analysis (EDA) was performed to understand the structure of the mutual fund datasets, identify patterns, detect anomalies, and generate initial insights before conducting advanced analytics. The analysis focused on industry growth trends, fund characteristics, investor behavior, and market participation.

EDA plays a crucial role in financial analytics as it helps uncover hidden relationships within the data and provides a foundation for performance evaluation and dashboard development.

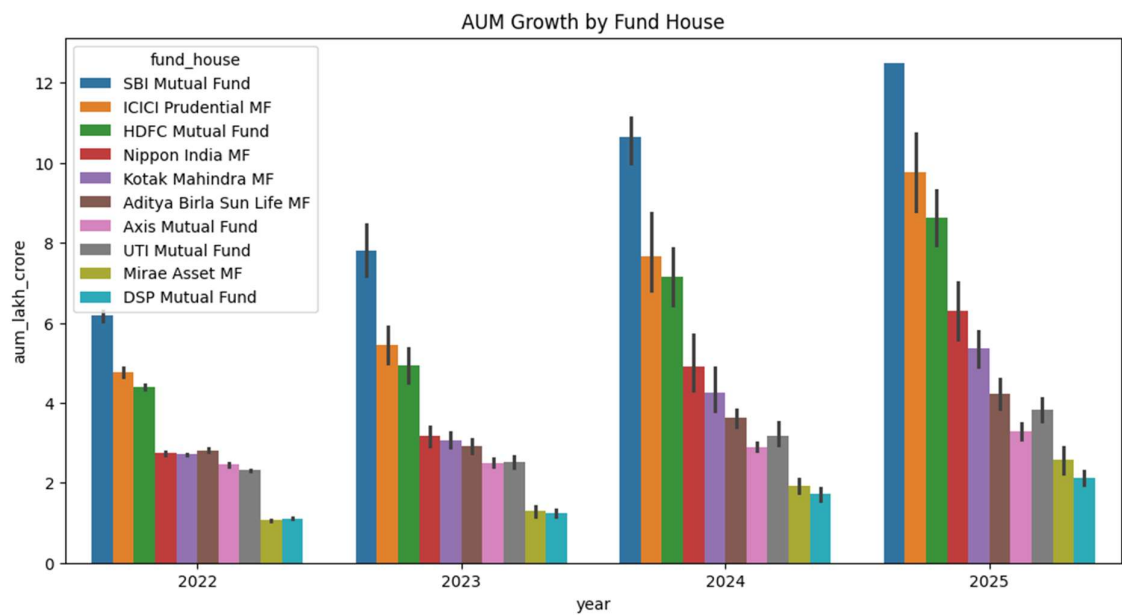
Industry AUM Analysis

Assets Under Management (AUM) represent the total market value of assets managed by mutual fund companies. AUM is an important indicator of industry growth, investor confidence, and market participation.

The analysis of industry AUM data from 2022 to 2025 revealed a consistent upward trend. The growth in AUM indicates increasing investor participation in mutual funds and a shift towards long-term investment strategies.

Key Observations

- Industry AUM increased significantly during the analysis period.
- Equity-oriented schemes attracted a major share of investments.
- Market growth and SIP participation contributed to rising AUM.
- Mutual funds continued to gain popularity among retail investors.



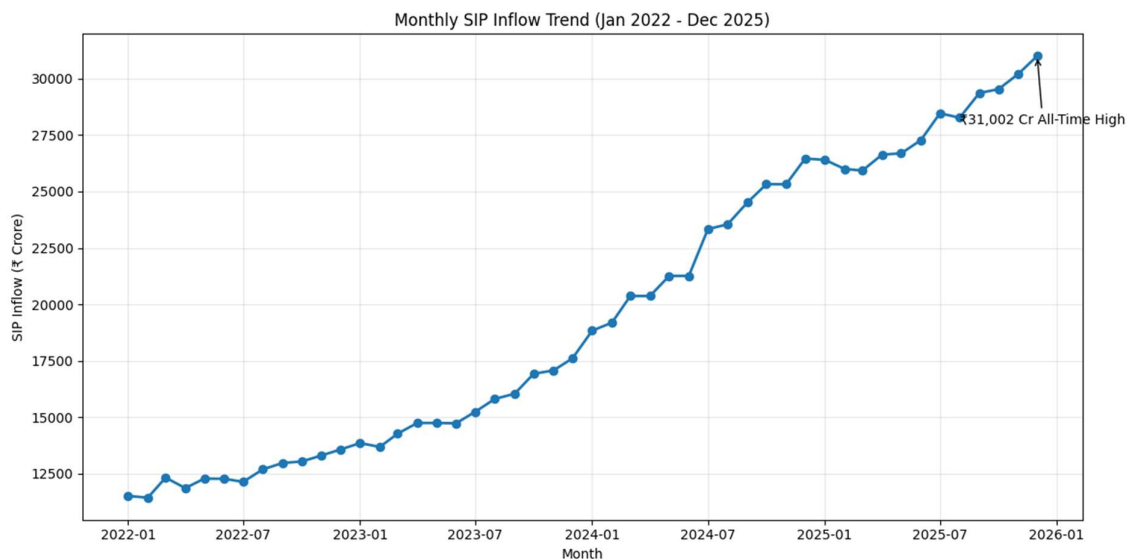
SIP Inflow Analysis

Systematic Investment Plans (SIPs) have become one of the most preferred investment methods among retail investors. Monthly SIP inflow data was analyzed to understand investor commitment and participation trends.

The data showed a steady increase in SIP contributions over time, indicating growing financial awareness and long-term investment behavior among investors.

Key Observations

- Monthly SIP inflows demonstrated a positive trend.
- Retail investor participation increased consistently.
- SIPs contributed significantly to industry growth.
- Periodic market volatility did not substantially affect SIP participation.



Fund Category Analysis

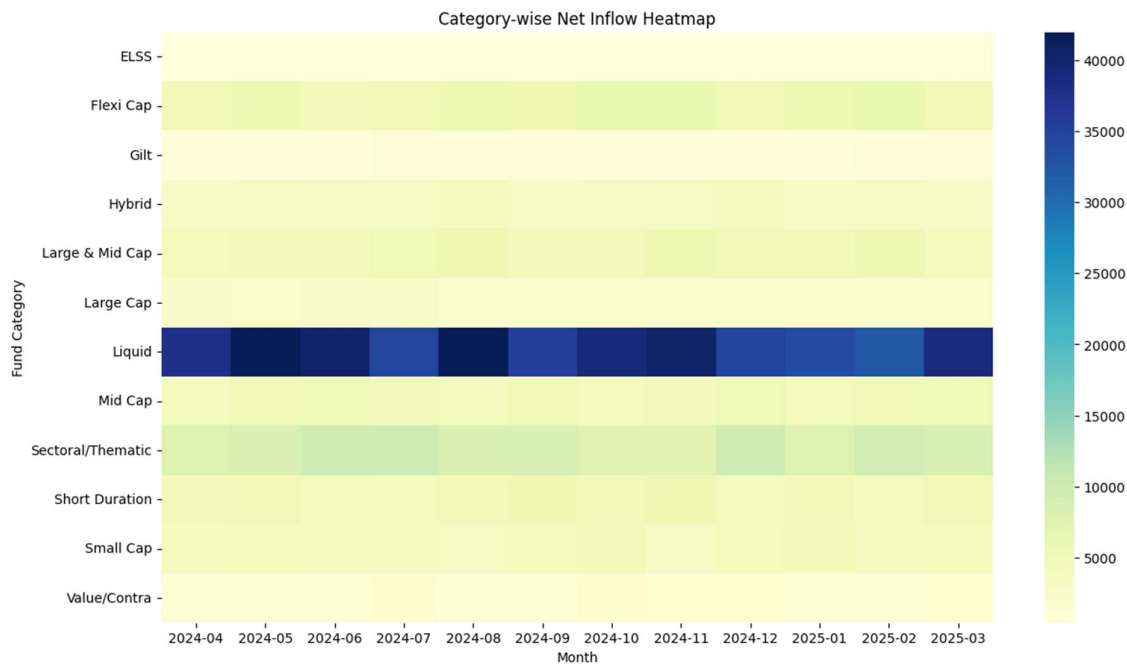
Mutual fund schemes were categorized into different investment categories including equity, debt, hybrid, and sector-specific funds.

The analysis highlighted that equity funds represented the largest share of schemes and assets under management. This reflects investors' preference for growth-oriented investments over long-term horizons.

Key Observations

- Equity funds dominated the mutual fund landscape.
- Hybrid funds provided balanced risk-return characteristics.
- Debt funds attracted relatively conservative investors.

- Category diversification allows investors to choose funds based on risk appetite.



Fund House Analysis

Fund houses (AMCs) were analyzed to understand market concentration and competitive positioning within the industry.

Several AMCs accounted for a substantial portion of total assets under management. Large fund houses benefited from strong brand recognition, diversified product offerings, and established investor trust.

Key Observations

- A small number of AMCs managed a large share of industry assets.
- Larger AMCs demonstrated stronger market presence.
- Competition remains intense among leading fund houses.
- Brand reputation plays an important role in attracting investors.

Summary of EDA Findings

The exploratory analysis provided several important insights regarding industry growth, investor participation, and mutual fund characteristics. The consistent increase in AUM and SIP inflows indicates a healthy and expanding mutual fund industry. Equity-oriented schemes continue to dominate investor preference, while leading AMCs maintain significant market share. These findings served as the foundation for subsequent performance analytics, risk assessment, and dashboard development stages of the project.

Performance Analytics

Introduction

Performance analytics was conducted to evaluate the effectiveness of mutual fund schemes using both return-based and risk-adjusted performance metrics. While absolute returns provide information about growth, they do not account for the level of risk taken by a fund. Therefore, multiple financial indicators were calculated to provide a comprehensive assessment of fund quality.

The analysis included Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. These metrics helped identify funds that generated superior returns while effectively managing risk.

Compound Annual Growth Rate (CAGR)

CAGR measures the average annual growth rate of an investment over a specified period, assuming profits are reinvested. It provides a standardized measure of performance and enables comparison across funds with different growth patterns.

The analysis calculated CAGR for 1-year, 3-year, and 5-year periods for all schemes.

Key Observations

- Several equity funds delivered strong long-term growth.
- Five-year CAGR provided a better indication of consistency than short-term returns.
- Some funds with high short-term returns demonstrated lower long-term stability.
- Long-term investors benefited significantly from compounding effects.

Sharpe Ratio Analysis

The Sharpe Ratio evaluates risk-adjusted returns by measuring the excess return generated for each unit of risk taken.

Funds with higher Sharpe Ratios are considered more efficient because they deliver superior returns relative to volatility.

Key Observations

- High-performing funds consistently achieved higher Sharpe Ratios.
- Some funds generated strong returns but exhibited excessive volatility.
- Risk-adjusted rankings differed from return-based rankings.
- Sharpe Ratio proved useful for identifying efficient investment options.

Alpha and Beta Analysis

Alpha measures a fund's ability to outperform its benchmark, while Beta measures sensitivity to market movements.

A positive Alpha indicates that the fund generated returns beyond what would be expected based on market performance. Beta values greater than one indicate higher market sensitivity, while values below one indicate lower volatility relative to the benchmark.

Key Observations

- Several funds generated positive Alpha, indicating superior management performance.
- Funds with lower Beta values exhibited relatively stable performance.
- High-Beta funds experienced greater fluctuations during market movements.
- Alpha and Beta together provided valuable insights into fund risk-return characteristics.

Sortino Ratio Analysis

The Sortino Ratio focuses specifically on downside risk rather than overall volatility. This makes it particularly useful for evaluating investment strategies where minimizing losses is a priority.

Compared with the Sharpe Ratio, the Sortino Ratio offers a more focused view of risk-adjusted performance by penalizing only negative volatility.

Key Observations

- Top-performing funds maintained favorable Sortino Ratios.
- Downside risk varied significantly across categories.
- Some funds demonstrated strong downside protection despite moderate returns.
- Sortino Ratio complemented Sharpe Ratio in evaluating risk efficiency.

Maximum Drawdown Analysis

Maximum Drawdown measures the largest decline from a historical peak to a subsequent trough. It provides insight into potential losses that investors may experience during adverse market conditions.

Funds with lower drawdowns are generally considered more resilient during market downturns.

Key Observations

- Maximum drawdown varied across categories and schemes.
- Aggressive funds experienced larger declines during volatile periods.

- Diversified funds generally exhibited lower drawdowns.
- Drawdown analysis highlighted the importance of risk management.

Fund Scorecard and Ranking

To simplify fund comparison, a composite fund score was created using multiple performance and risk metrics. The scoring methodology considered returns, Sharpe Ratio, Alpha, expense ratio, and drawdown characteristics.

This approach enabled objective ranking of mutual fund schemes and helped identify the strongest overall performers.

Key Observations

- Top-ranked funds combined strong returns with effective risk management.
- High scores were generally associated with superior Sharpe Ratios.
- Funds with excessive drawdowns received lower rankings.
- Composite scoring improved overall fund comparison.

Summary

Performance analytics demonstrated that evaluating mutual funds solely on returns can be misleading. Risk-adjusted metrics such as Sharpe Ratio, Sortino Ratio, Alpha, and Maximum Drawdown provide a more comprehensive view of fund quality. The results showed that several funds consistently outperformed peers while maintaining acceptable risk levels, making them attractive options for long-term investors.

Dashboard Development and Visualization

Introduction

One of the primary objectives of this project was to convert complex mutual fund datasets into an interactive and user-friendly dashboard. To achieve this, Microsoft Power BI was used to create a multi-page dashboard that provides insights into industry growth, fund performance, investor behavior, and SIP trends.

The dashboard enables users to interact with data through filters, drill-through functionality, and dynamic visualizations. This approach allows investors, analysts, and decision-makers to explore mutual fund performance and industry trends efficiently.

The dashboard was designed using the Bluestock theme and organized into four major sections.

Dashboard Page 1: Industry Overview

The Industry Overview page provides a high-level summary of the Indian mutual fund industry.

Visual Components

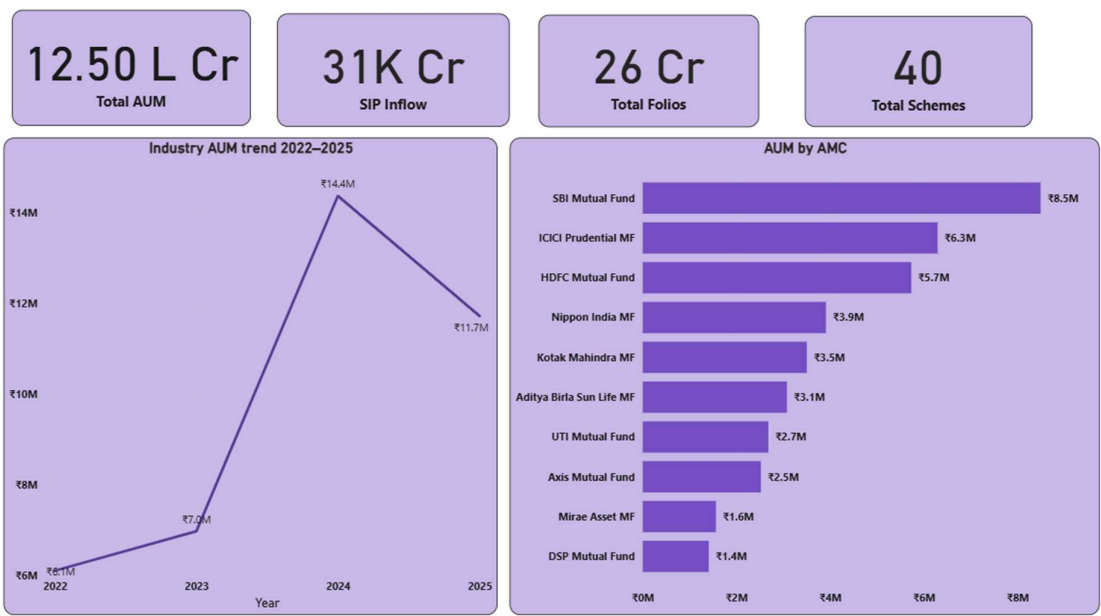
- KPI Card: Total Industry AUM
- KPI Card: Monthly SIP Inflow
- KPI Card: Total Folios
- KPI Card: Total Schemes
- Industry AUM Trend Line Chart
- AUM by AMC Bar Chart

Purpose

The objective of this page is to provide a quick overview of industry growth and market participation. Investors can immediately understand the scale of the industry and observe long-term trends.

Key Insights

- Industry AUM demonstrated continuous growth.
- SIP contributions remained strong throughout the analysis period.
- Leading AMCs controlled a significant portion of industry assets.
- Investor participation continued to increase.



Dashboard Page 2: Fund Performance

The Fund Performance page focuses on evaluating mutual fund schemes using risk and return metrics.

Visual Components

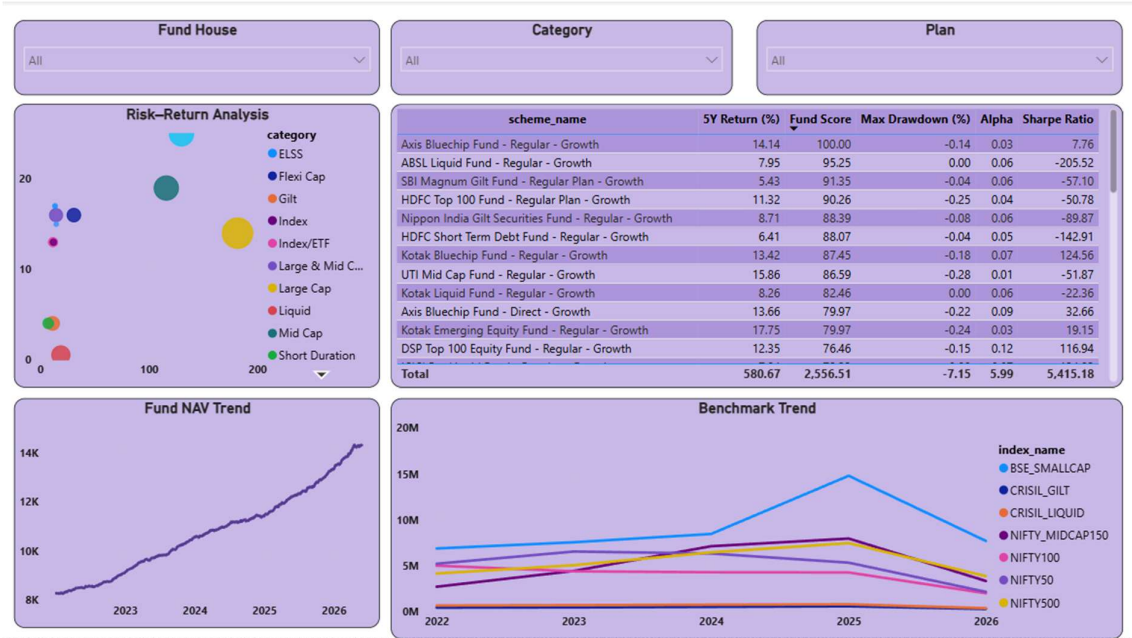
- Risk vs Return Scatter Plot
- Fund Scorecard Table
- NAV Trend Visualization
- Benchmark Comparison
- Fund House Slicer
- Category Slicer
- Plan Slicer

Purpose

This page enables users to compare schemes based on returns, volatility, and overall performance. It supports informed investment decisions through visual comparison.

Key Insights

- High-return funds exhibited varying levels of risk.
- Some funds delivered superior risk-adjusted performance.
- Benchmark comparisons highlighted outperforming schemes.
- Fund score rankings simplified fund selection.



Dashboard Page 3: Investor Analytics

The Investor Analytics page examines investor behavior and transaction patterns.

Visual Components

- State-wise Investment Distribution
- Transaction Type Distribution
- Age Group Analysis
- Monthly Transaction Trends
- Geographic Filters

Purpose

This page helps identify investment preferences across demographic and geographic segments.

Key Insights

- Certain states contributed a larger share of investment activity.
- SIP remained the dominant investment mode.
- Younger investors demonstrated growing participation.
- Transaction activity increased over time.



Dashboard Page 4: SIP and Market Trends

The SIP and Market Trends page combines SIP inflow analysis with market performance indicators.

Visual Components

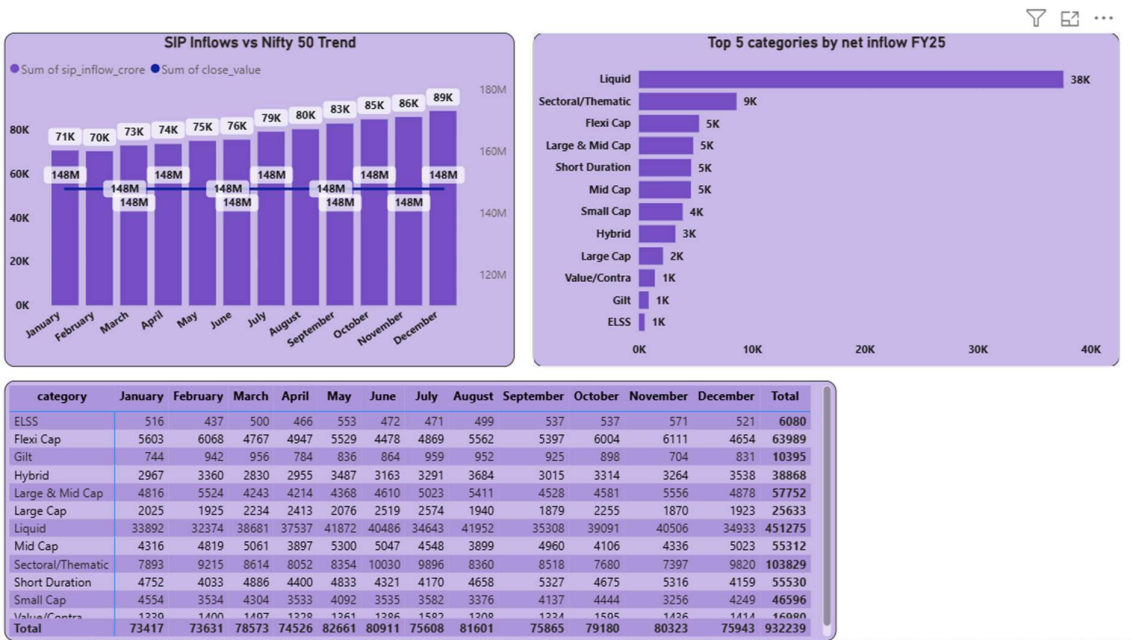
- SIP Inflow vs Nifty Trend Chart
- Category Inflow Analysis
- Top Categories by Inflow
- Market Trend Indicators

Purpose

This page provides insight into how investor behavior aligns with market conditions.

Key Insights

- SIP inflows continued to rise despite market fluctuations.
- Equity-oriented categories attracted significant investments.
- Investor confidence remained strong during volatile periods.
- Long-term investment behavior was evident across categories.



Dashboard Features

Several advanced Power BI features were incorporated into the dashboard:

Drill-Through Functionality

Users can navigate from summary views to detailed fund-level information.

Interactive Filters

Slicers allow filtering by:

- Fund House
- Category
- Plan Type
- State
- Age Group

Tooltips

Additional information is displayed when users hover over visual elements.

Dynamic Cross-Filtering

Selecting one visual automatically updates related visuals across the dashboard.

Dashboard Benefits

The dashboard transforms large volumes of mutual fund data into actionable insights. It enables users to:

- Monitor industry growth.
- Compare fund performance.
- Analyze investor behavior.
- Track SIP participation trends.
- Support investment decision-making.

The dashboard serves as the visual layer of the analytics framework and demonstrates how business intelligence tools can enhance financial analysis.

Limitations

Despite providing comprehensive insights into mutual fund performance and investor behavior, the project has certain limitations.

1. The analysis is based on historical data and therefore cannot guarantee future fund performance.
2. Some datasets were simulated or limited in scope and may not fully represent real-world market conditions.
3. The fund recommendation engine uses a simple risk-based approach and does not consider factors such as investor age, income, investment horizon, or financial goals.
4. Portfolio holdings data represents a snapshot of fund composition and may not capture frequent changes made by fund managers.
5. The project does not incorporate real-time market data, which may affect the timeliness of certain analyses.
6. External economic factors such as inflation, interest rate changes, geopolitical events, and market sentiment were not included in the analysis.
7. Advanced machine learning techniques for return forecasting and investor segmentation were outside the scope of this project.

Recommendations and Future Scope

Based on the findings of this project, several recommendations can be made to improve mutual fund analysis and investor decision-making.

1. Investors should evaluate funds using risk-adjusted metrics such as Sharpe Ratio, Alpha, and Maximum Drawdown rather than relying solely on historical returns.
2. Fund houses should monitor SIP continuity and identify at-risk investors to improve retention and reduce SIP discontinuation rates.
3. Portfolio diversification should be regularly assessed to avoid excessive concentration in specific sectors or securities.
4. Real-time NAV and market data integration can enhance the accuracy and usefulness of the analytics platform.
5. The fund recommendation engine can be improved by incorporating investor demographics, financial goals, investment horizon, and risk tolerance.
6. Machine learning models can be implemented to predict fund performance, investor churn, and future SIP trends.
7. Future versions of the dashboard can include live market tracking, automated alerts, and personalized investment recommendations.
8. Cloud deployment and web-based dashboard integration can improve accessibility and scalability for a larger user base.

Overall, the project demonstrates the potential of data analytics and business intelligence tools in supporting informed investment decisions and improving mutual fund evaluation processes.

Key Findings

1. The mutual fund industry demonstrated strong growth in Assets Under Management (AUM) and SIP inflows during the analysis period.
2. Risk-adjusted performance metrics such as Sharpe Ratio and Sortino Ratio provided better fund evaluation than absolute returns alone.
3. Several funds consistently generated positive Alpha, indicating outperformance relative to benchmark indices.
4. Historical VaR and CVaR analysis highlighted differences in downside risk across schemes.
5. Investor behavior analysis revealed the importance of SIP continuity in long-term wealth creation.
6. Sector concentration analysis showed that most equity funds maintained diversified portfolios with moderate concentration levels.
7. The Power BI dashboard successfully transformed complex financial data into actionable business insights.
8. The risk-based recommendation engine demonstrated the practical application of analytics in investment decision-making.

Conclusion

The Bluestock Mutual Fund Analytics project successfully developed an end-to-end analytics framework for evaluating mutual fund performance, risk characteristics, investor behavior, and industry trends. The project integrated multiple datasets and applied data engineering, statistical analysis, business intelligence, and financial analytics techniques to generate meaningful insights.

Through performance analytics, advanced risk assessment, investor analytics, and dashboard development, the project demonstrated how data-driven decision-making can improve mutual fund evaluation and investment analysis. The Power BI dashboard provided an interactive platform for visualizing industry trends and fund performance, while the recommendation engine showcased the practical application of analytical models.

Overall, the project achieved all defined objectives and delivered a scalable analytics solution that can be extended with real-time data integration, predictive analytics, and machine learning capabilities in future implementations.

References

1. Association of Mutual Funds in India (AMFI)
2. National Stock Exchange (NSE) India
3. Securities and Exchange Board of India (SEBI)
4. Microsoft Power BI Documentation
5. Python Documentation
6. Pandas Documentation
7. NumPy Documentation
8. Plotly Documentation
9. SQLite Documentation
10. Financial Risk Management Literature and Industry Reports